# Ablation : ConvNeXt-Tiny — SiLU | Drop | BatchNorm

**Config:** ImageNet- K pretrained | SiLU activation | Dropout | BatchNorm **Model:** ConvNeXt-Tir

Dataset: Rice Leaf Disease Identification Dataset

## . Colab Setup (jalankan sekali per session)

In [1]:
```
from google.colab import drive
drive.mount('/content/drive')
import os
DRIVE_DIR = '/content/drive/MyDrive/rice-convnext/08_SiLU_Dropout_BatchNo
os.makedirs(DRIVE_DIR, exist_ok=True)
print(f'Drive folder ready: {DRIVE_DIR}')
```

```
Mounted at /content/drive
Drive folder ready: /content/drive/MyDrive/rice-convnext/08_SiLU_Dropout_Batc
```

In [2]:
```
from google.colab import files
import os, shutil
print('Upload kaggle.json...')
files.upload()
os.makedirs('/root/.kaggle', exist_ok=True)
shutil.move('kaggle.json', '/root/.kaggle/kaggle.json')
os.chmod('/root/.kaggle/kaggle.json', 0o600)
print('Kaggle API ready!')
```

```
Upload kaggle.json...
```

Upload widget is only available when the cell has been executed in the current browser session. Pl rerun this cell to enable.

```
Saving kaggle.json to kaggle.json
Kaggle API ready!
```

In [3]:
```
DATASET_DIR = '/content/drive/MyDrive/rice-convnext/dataset'
if os.path.exists(DATASET_DIR) and len(os.listdir(DATASET_DIR)) > 0:
    print('Dataset sudah ada, skip download.')
else:
    !pip install kaggle -q
    !kaggle datasets download -d wangxiaoqii/rice-leaf-disease-identifica
    !unzip -q /content/tmp/*.zip -d {DATASET_DIR}
    !rm -rf /content/tmp
    print('Dataset ready!')

import shutil
LOCAL_DATASET = '/content/dataset'
if not os.path.exists(LOCAL_DATASET):
    print('Copying dataset ke local SSD...')
    shutil.copytree(DATASET_DIR, LOCAL_DATASET)
    print('Done!')
else:
```

```
    print('Sudah ada di local.')
DATA_DIR = LOCAL_DATASET
```

```
Dataset sudah ada, skip download.
Copying dataset ke local SSD...
Done!
```

## . Setup & Imports

In [4]:
```
import os, random, shutil
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from collections import Counter
from sklearn.metrics import classification_report, confusion_matrix
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader, random_split
from torchvision import datasets, transforms, models

SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
torch.cuda.manual_seed(SEED)
torch.backends.cudnn.deterministic = True

DEVICE = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Device: {DEVICE}')
if torch.cuda.is_available():
    print(f'GPU: {torch.cuda.get_device_name(0)}')
```

```
Device: cuda
GPU: NVIDIA A100-SXM4-80GB
```

## . Configuration

In [5]:
```
DATA_DIR      = '/content/dataset'
DRIVE_DIR     = '/content/drive/MyDrive/rice-convnext/08_SiLU_Dropout_Bat
IMG_SIZE      = 224
BATCH_SIZE    = 64
NUM_EPOCHS    = 30
LR_INIT       = 1e-3
LR_FINAL      = 1e-6
WEIGHT_DECAY  = 0.05
NUM_WORKERS   = 2
IMAGENET_MEAN = [0.485, 0.456, 0.406]
IMAGENET_STD  = [0.229, 0.224, 0.225]
SAVE_PATH     = f'{DRIVE_DIR}/convnext_tiny_08_silu_dropout_batchnorm.pt'
EXP_NAME      = 'ConvNeXt-Tiny | SiLU | Dropout | BatchNorm'
print(f'Experiment: {EXP_NAME}')
```

```
Experiment: ConvNeXt-Tiny | SiLU | Dropout | BatchNorm
```

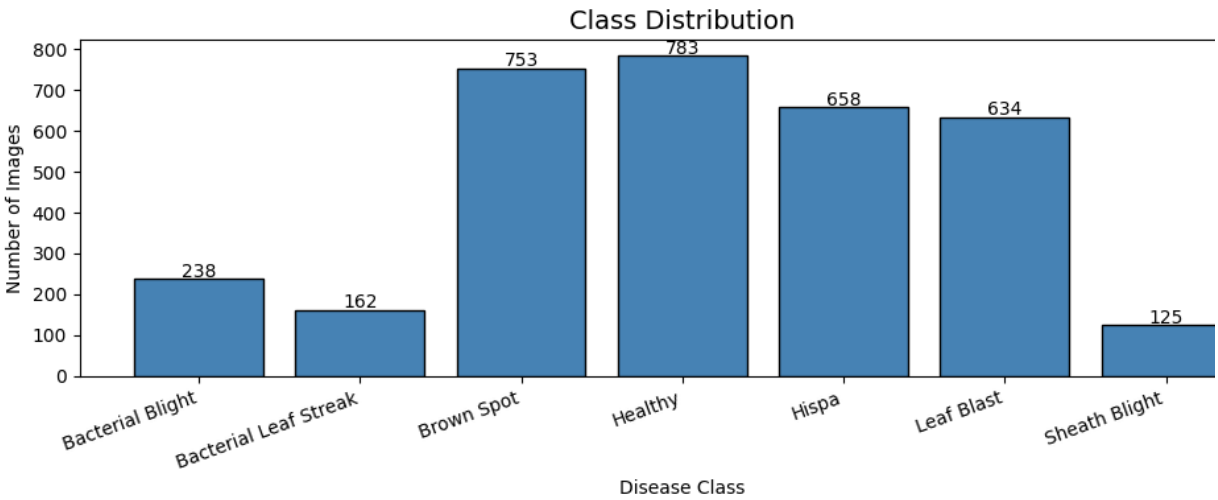
## . Data Loading & Augmentation

In [6]:
```
train_transforms = transforms.Compose([
    transforms.RandomResizedCrop(IMG_SIZE),
    transforms.RandomHorizontalFlip(),
    transforms.RandomVerticalFlip(),
    transforms.RandomRotation(30),
    transforms.ColorJitter(brightness=0.2, contrast=0.2, saturation=0.2),
    transforms.ToTensor(),
    transforms.Normalize(mean=IMAGENET_MEAN, std=IMAGENET_STD),
])
val_test_transforms = transforms.Compose([
    transforms.Resize(256),
    transforms.CenterCrop(IMG_SIZE),
    transforms.ToTensor(),
    transforms.Normalize(mean=IMAGENET_MEAN, std=IMAGENET_STD),
])

full_dataset = datasets.ImageFolder(DATA_DIR)
CLASS_NAMES  = full_dataset.classes
NUM_CLASSES  = len(CLASS_NAMES)
print(f'Classes ({NUM_CLASSES}): {CLASS_NAMES}')
print(f'Total images: {len(full_dataset)}')

total   = len(full_dataset)
n_train = int(0.6 * total)
n_val   = int(0.2 * total)
n_test  = total - n_train - n_val

generator = torch.Generator().manual_seed(SEED)
train_set, val_set, test_set = random_split(
    full_dataset, [n_train, n_val, n_test], generator=generator
)

train_set.dataset.transform = train_transforms
val_set.dataset.transform   = val_test_transforms
test_set.dataset.transform  = val_test_transforms

train_loader = DataLoader(train_set, batch_size=BATCH_SIZE, shuffle=True,
val_loader   = DataLoader(val_set,   batch_size=BATCH_SIZE, shuffle=False
test_loader  = DataLoader(test_set,  batch_size=BATCH_SIZE, shuffle=False
print(f'Train: {len(train_set)} | Val: {len(val_set)} | Test: {len(test_s
```

```
Classes (7): ['Bacterial Blight', 'Bacterial Leaf Streak', 'Brown Spot', 'Hea
'Hispa', 'Leaf Blast', 'Sheath Blight']
Total images: 3353
Train: 2011 | Val: 670 | Test: 672
```

In [7]:
```
labels_all = [full_dataset.targets[i] for i in range(len(full_dataset))]
counts = [Counter(labels_all)[i] for i in range(NUM_CLASSES)]
plt.figure(figsize=(10, 4))
bars = plt.bar(CLASS_NAMES, counts, color='steelblue', edgecolor='black')
plt.title('Class Distribution', fontsize=14)
plt.xlabel('Disease Class'); plt.ylabel('Number of Images')
plt.xticks(rotation=20, ha='right')
for bar, count in zip(bars, counts):
    plt.text(bar.get_x() + bar.get_width()/2, bar.get_height()+5, str(cou
plt.tight_layout()
```

```
plt.savefig('class_distribution.png', dpi=150)
plt.show()
```




## . Model — ConvNeXt-Tiny (SiLU | Dropout | BatchNor

In [8]:
```
class BatchNormNHWC(nn.Module):
    """BatchNorm2d wrapper that accepts NHWC input (used inside ConvNeXt
    def __init__(self, num_features):
        super().__init__()
        self.bn = nn.BatchNorm2d(num_features)
    def forward(self, x):  # x: (B, H, W, C)
        return self.bn(x.permute(0, 3, 1, 2)).permute(0, 2, 3, 1)

def replace_layernorm_with_batchnorm(module):
    """Recursively replace all LayerNorm variants with BatchNorm."""
    for name, child in module.named_children():
        if type(child).__name__ == 'LayerNorm2d':
            # Stem / downsampling: NCHW input → use BatchNorm2d
            C = child.normalized_shape[0]
            setattr(module, name, nn.BatchNorm2d(C))
        elif isinstance(child, nn.LayerNorm):
            # CNBlock (NHWC after Permute) → use BatchNormNHWC
            C = child.normalized_shape[0]
            setattr(module, name, BatchNormNHWC(C))
        else:
            replace_layernorm_with_batchnorm(child)

def replace_activation(model, src_cls, dst_fn):
    for name, module in model.named_children():
        if isinstance(module, src_cls):
            setattr(model, name, dst_fn())
        else:
            replace_activation(module, src_cls, dst_fn)

from torchvision.ops import StochasticDepth

def replace_stochastic_depth(model, p=0.1):
    for name, module in model.named_children():
        if isinstance(module, StochasticDepth):
            setattr(model, name, nn.Dropout(p=p))
        else:
            replace_stochastic_depth(module, p)
```

```
model = models.convnext_tiny(weights=models.ConvNeXt_Tiny_Weights.IMAGENE
in_features = model.classifier[2].in_features
model.classifier[2] = nn.Linear(in_features, NUM_CLASSES)

# Replace GELU → SiLU
replace_activation(model, nn.GELU, nn.SiLU)
print('Activation: GELU → SiLU')

# Replace Stochastic Depth → Dropout(p=0.1)
replace_stochastic_depth(model, p=0.1)
print('Regularization: StochasticDepth → Dropout(p=0.1)')

# Replace LayerNorm → BatchNorm
replace_layernorm_with_batchnorm(model)
# Fix classifier: after Flatten input is (B, C), use BatchNorm1d
if isinstance(model.classifier[2], BatchNormNHWC):
    C = model.classifier[2].bn.num_features
    model.classifier[2] = nn.BatchNorm1d(C)
print('Normalization: LayerNorm → BatchNorm')

model = model.to(DEVICE)
total_p     = sum(p.numel() for p in model.parameters())
trainable_p = sum(p.numel() for p in model.parameters() if p.requires_gra
print(f'Experiment : {EXP_NAME}')
print(f'Total params    : {total_p:,}')
print(f'Trainable params: {trainable_p:,}')
```

```
Downloading: "https://download.pytorch.org/models/convnext_tiny-983f1562.pth"
root/.cache/torch/hub/checkpoints/convnext_tiny-983f1562.pth
100%|██████████| 109M/109M [00:00<00:00, 184MB/s]
Activation: GELU → SiLU
Regularization: StochasticDepth → Dropout(p=0.1)
Normalization: LayerNorm → BatchNorm
Experiment : ConvNeXt-Tiny | SiLU | Dropout | BatchNorm
Total params    : 27,825,511
Trainable params: 27,825,511
```

## . Training Setup

In [9]:
```
criterion = nn.CrossEntropyLoss()
optimizer = optim.AdamW(model.parameters(), lr=LR_INIT, weight_decay=WEIG

def get_scheduler(optimizer, num_epochs, lr_init, lr_final):
    warmup = optim.lr_scheduler.LinearLR(optimizer, start_factor=0.01, en
    cosine = optim.lr_scheduler.CosineAnnealingLR(optimizer, T_max=num_ep
    return optim.lr_scheduler.SequentialLR(optimizer, schedulers=[warmup,

scheduler = get_scheduler(optimizer, NUM_EPOCHS, LR_INIT, LR_FINAL)
```

## . Training Loop

In [10]:
```
def train_one_epoch(model, loader, criterion, optimizer):
    model.train()
    running_loss, correct, total = 0.0, 0, 0
    for images, labels in loader:
        images, labels = images.to(DEVICE), labels.to(DEVICE)
        optimizer.zero_grad()
```

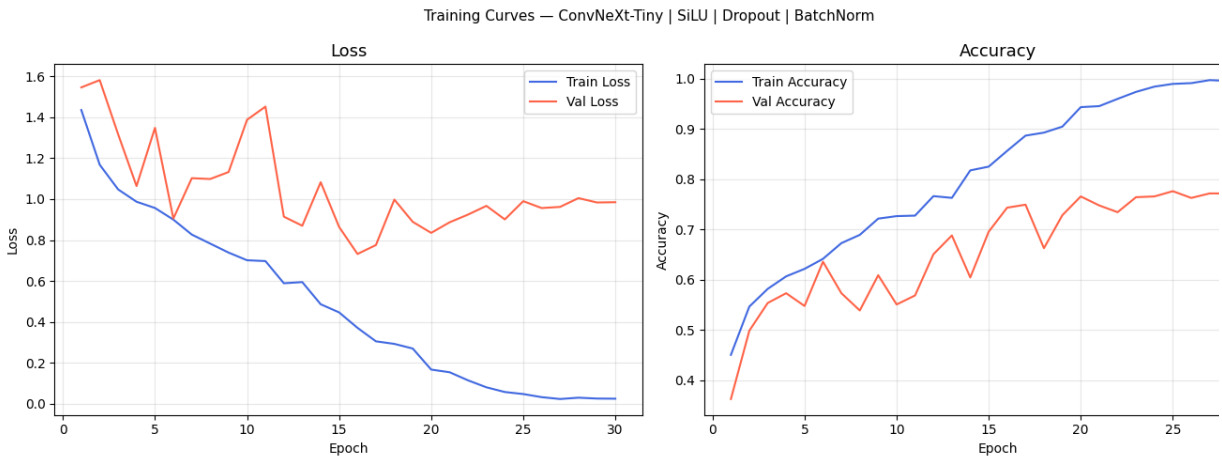
```
        outputs = model(images)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()
        running_loss += loss.item() * images.size(0)
        _, predicted = outputs.max(1)
        correct += predicted.eq(labels).sum().item()
        total   += labels.size(0)
    return running_loss / total, correct / total

def evaluate(model, loader, criterion):
    model.eval()
    running_loss, correct, total = 0.0, 0, 0
    with torch.no_grad():
        for images, labels in loader:
            images, labels = images.to(DEVICE), labels.to(DEVICE)
            outputs = model(images)
            loss = criterion(outputs, labels)
            running_loss += loss.item() * images.size(0)
            _, predicted = outputs.max(1)
            correct += predicted.eq(labels).sum().item()
            total   += labels.size(0)
    return running_loss / total, correct / total

history = {'train_loss': [], 'train_acc': [], 'val_loss': [], 'val_acc':
best_val_acc = 0.0

for epoch in range(NUM_EPOCHS):
    train_loss, train_acc = train_one_epoch(model, train_loader, criterio
    val_loss,   val_acc   = evaluate(model, val_loader, criterion)
    scheduler.step()
    history['train_loss'].append(train_loss)
    history['train_acc'].append(train_acc)
    history['val_loss'].append(val_loss)
    history['val_acc'].append(val_acc)
    if val_acc > best_val_acc:
        best_val_acc = val_acc
        torch.save(model.state_dict(), SAVE_PATH)
    print(f'Epoch [{epoch+1:02d}/{NUM_EPOCHS}] '
          f'Train Loss: {train_loss:.4f} Acc: {train_acc:.4f} | '
          f'Val Loss: {val_loss:.4f} Acc: {val_acc:.4f}'
          + (' <- best' if val_acc == best_val_acc else ''))

print(f'\nBest Val Accuracy: {best_val_acc:.4f}')
print(f'Saved: {SAVE_PATH}')
```

```
Epoch [01/30] Train Loss: 1.4362 Acc: 0.4505 | Val Loss: 1.5468 Acc: 0.3627 <
Epoch [02/30] Train Loss: 1.1683 Acc: 0.5465 | Val Loss: 1.5823 Acc: 0.4985 <
Epoch [03/30] Train Loss: 1.0476 Acc: 0.5818 | Val Loss: 1.3175 Acc: 0.5537 <
Epoch [04/30] Train Loss: 0.9873 Acc: 0.6067 | Val Loss: 1.0641 Acc: 0.5731 <
Epoch [05/30] Train Loss: 0.9567 Acc: 0.6216 | Val Loss: 1.3487 Acc: 0.5478
Epoch [06/30] Train Loss: 0.8998 Acc: 0.6415 | Val Loss: 0.9051 Acc: 0.6358 <
Epoch [07/30] Train Loss: 0.8266 Acc: 0.6728 | Val Loss: 1.1025 Acc: 0.5731
Epoch [08/30] Train Loss: 0.7825 Acc: 0.6892 | Val Loss: 1.0986 Acc: 0.5388
Epoch [09/30] Train Loss: 0.7384 Acc: 0.7215 | Val Loss: 1.1328 Acc: 0.6090
Epoch [10/30] Train Loss: 0.7010 Acc: 0.7265 | Val Loss: 1.3883 Acc: 0.5507
Epoch [11/30] Train Loss: 0.6970 Acc: 0.7275 | Val Loss: 1.4528 Acc: 0.5687
Epoch [12/30] Train Loss: 0.5884 Acc: 0.7663 | Val Loss: 0.9143 Acc: 0.6507 <
Epoch [13/30] Train Loss: 0.5941 Acc: 0.7628 | Val Loss: 0.8698 Acc: 0.6881 <
Epoch [14/30] Train Loss: 0.4856 Acc: 0.8175 | Val Loss: 1.0830 Acc: 0.6045
Epoch [15/30] Train Loss: 0.4459 Acc: 0.8250 | Val Loss: 0.8640 Acc: 0.6955 <
Epoch [16/30] Train Loss: 0.3701 Acc: 0.8563 | Val Loss: 0.7317 Acc: 0.7433 <
Epoch [17/30] Train Loss: 0.3042 Acc: 0.8866 | Val Loss: 0.7754 Acc: 0.7493 <
Epoch [18/30] Train Loss: 0.2917 Acc: 0.8926 | Val Loss: 0.9978 Acc: 0.6627
Epoch [19/30] Train Loss: 0.2690 Acc: 0.9045 | Val Loss: 0.8888 Acc: 0.7284
Epoch [20/30] Train Loss: 0.1660 Acc: 0.9433 | Val Loss: 0.8350 Acc: 0.7657 <
Epoch [21/30] Train Loss: 0.1530 Acc: 0.9453 | Val Loss: 0.8866 Acc: 0.7478
Epoch [22/30] Train Loss: 0.1134 Acc: 0.9597 | Val Loss: 0.9244 Acc: 0.7343
Epoch [23/30] Train Loss: 0.0792 Acc: 0.9736 | Val Loss: 0.9671 Acc: 0.7642
Epoch [24/30] Train Loss: 0.0562 Acc: 0.9841 | Val Loss: 0.9008 Acc: 0.7657 <
Epoch [25/30] Train Loss: 0.0463 Acc: 0.9896 | Val Loss: 0.9901 Acc: 0.7761 <
Epoch [26/30] Train Loss: 0.0311 Acc: 0.9910 | Val Loss: 0.9564 Acc: 0.7627
Epoch [27/30] Train Loss: 0.0220 Acc: 0.9970 | Val Loss: 0.9621 Acc: 0.7716
Epoch [28/30] Train Loss: 0.0285 Acc: 0.9955 | Val Loss: 1.0050 Acc: 0.7716
Epoch [29/30] Train Loss: 0.0242 Acc: 0.9965 | Val Loss: 0.9837 Acc: 0.7627
Epoch [30/30] Train Loss: 0.0236 Acc: 0.9960 | Val Loss: 0.9850 Acc: 0.7701

Best Val Accuracy: 0.7761
Saved: /content/drive/MyDrive/rice-convnext/08_SiLU_Dropout_BatchNorm/
convnext_tiny_08_silu_dropout_batchnorm.pt
```

## . Training Curves

In [11]:

```
epochs_range = range(1, NUM_EPOCHS + 1)
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].plot(epochs_range, history['train_loss'], label='Train Loss', col
axes[0].plot(epochs_range, history['val_loss'],   label='Val Loss',   col
axes[0].set_title('Loss', fontsize=13)
axes[0].set_xlabel('Epoch'); axes[0].set_ylabel('Loss')
axes[0].legend(); axes[0].grid(alpha=0.3)
axes[1].plot(epochs_range, history['train_acc'], label='Train Accuracy',
axes[1].plot(epochs_range, history['val_acc'],   label='Val Accuracy',
axes[1].set_title('Accuracy', fontsize=13)
axes[1].set_xlabel('Epoch'); axes[1].set_ylabel('Accuracy')
axes[1].legend(); axes[1].grid(alpha=0.3)
plt.suptitle(f'Training Curves — {EXP_NAME}', fontsize=11)
plt.tight_layout()
plt.savefig('training_curves.png', dpi=150)
plt.show()
```




## . Evaluation on Test Set

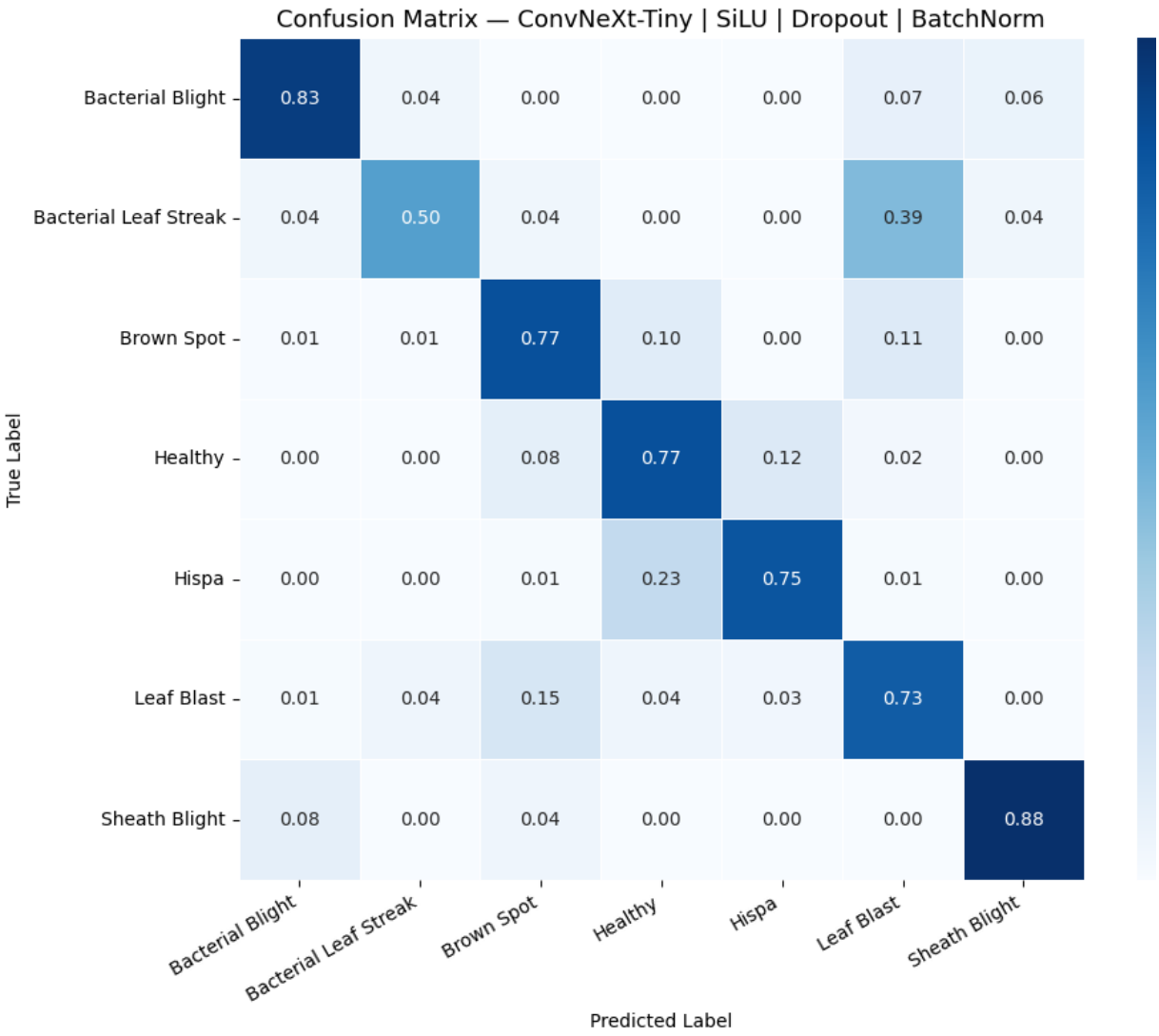
In [12]:
```
model.load_state_dict(torch.load(SAVE_PATH, map_location=DEVICE))
model.eval()
all_preds, all_labels = [], []
with torch.no_grad():
    for images, labels in test_loader:
        images = images.to(DEVICE)
        outputs = model(images)
        _, predicted = outputs.max(1)
        all_preds.extend(predicted.cpu().numpy())
        all_labels.extend(labels.numpy())

report = classification_report(all_labels, all_preds, target_names=CLASS_
report_df = pd.DataFrame(report).transpose()
print('=== Classification Report ===')
print(report_df.round(4).to_string())
report_df.to_csv('classification_report.csv')

summary = {
    'Model'     : ['ConvNeXt-Tiny'],
    'Activation': ['SiLU'],
    'Dropout'   : ['Dropout'],
    'Norm'      : ['BatchNorm'],
    'Accuracy'  : [report['accuracy']],
    'Precision' : [report['weighted avg']['precision']],
    'Recall'    : [report['weighted avg']['recall']],
    'F1-Score'  : [report['weighted avg']['f1-score']],
}
summary_df = pd.DataFrame(summary)
print('\n=== Summary ===')
print(summary_df.round(4).to_string(index=False))
summary_df.to_csv('summary.csv', index=False)
```

```
=== Classification Report ===
                      precision  recall  f1-score   support
Bacterial Blight         0.9000  0.8333    0.8654   54.0000
Bacterial Leaf Streak    0.6364  0.5000    0.5600   28.0000
Brown Spot               0.7853  0.7711    0.7781  166.0000
Healthy                  0.7314  0.7711    0.7507  166.0000
Hispa                    0.7757  0.7545    0.7650  110.0000
Leaf Blast               0.6977  0.7317    0.7143  123.0000
Sheath Blight            0.8462  0.8800    0.8627   25.0000
accuracy                 0.7589  0.7589    0.7589    0.7589
macro avg                0.7675  0.7488    0.7566  672.0000
weighted avg             0.7597  0.7589    0.7586  672.0000

=== Summary ===
        Model Activation Dropout      Norm  Accuracy  Precision  Recall  F1-S
ConvNeXt-Tiny       SiLU Dropout BatchNorm    0.7589     0.7597  0.7589    0.
```

## . Confusion Matrix

In [13]:

```python
cm = confusion_matrix(all_labels, all_preds)
cm_norm = cm.astype('float') / cm.sum(axis=1)[:, np.newaxis]
plt.figure(figsize=(10, 8))
sns.heatmap(cm_norm, annot=True, fmt='.2f',
            xticklabels=CLASS_NAMES, yticklabels=CLASS_NAMES,
            cmap='Blues', linewidths=0.5)
plt.title(f'Confusion Matrix — ConvNeXt-Tiny | SiLU | Dropout | BatchNorm
plt.ylabel('True Label'); plt.xlabel('Predicted Label')
plt.xticks(rotation=30, ha='right')
plt.tight_layout()
plt.savefig('confusion_matrix.png', dpi=150)
plt.show()
```




## . Save All to Drive

In [14]:
```
local_files = ['training_curves.png', 'confusion_matrix.png',
               'classification_report.csv', 'summary.csv', 'class_distrib
for fname in local_files:
    if os.path.exists(fname):
        shutil.copy(fname, f'{DRIVE_DIR}/{fname}')

all_outputs = [
    (f'{DRIVE_DIR}/convnext_tiny_08_silu_dropout_batchnorm.pt',
    (f'{DRIVE_DIR}/training_curves.png',              'Training curves'),
    (f'{DRIVE_DIR}/confusion_matrix.png',             'Confusion matrix')
    (f'{DRIVE_DIR}/classification_report.csv',        'Per-class metrics'
    (f'{DRIVE_DIR}/summary.csv',                      'Summary table'),
]
print(f'=== Output Files: {DRIVE_DIR} ===')
for fpath, desc in all_outputs:
    exists = '✅' if os.path.exists(fpath) else '❌'
    print(f'{exists}  {os.path.basename(fpath):<45} {desc}')
```

```
=== Output Files: /content/drive/MyDrive/rice-convnext/08_SiLU_Dropout_BatchN
===
✅  convnext_tiny_08_silu_dropout_batchnorm.pt    Model weights
✅  training_curves.png                           Training curves
✅  confusion_matrix.png                          Confusion matrix
✅  classification_report.csv                     Per-class metrics
✅  summary.csv                                   Summary table
```